SmartFlow NLEX — System Inventory

What is in the system now, module by module and feature by feature · as of 10 November 2026

> Read from the current working tree, including uncommitted changes. Nothing was run against the live database, so live figures are not re-checked. Items marked "not confirmed" were seen in code or comments but not verified against the stored data.

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1. System at a glance

- **Frontend:** Next.js (static export) on port 3002.
- **Backend:** Express / TypeScript API on port 4000.
- **Database:** AWS RDS Postgres, organised as bronze (raw), silver (cleaned) and gold (facts and model outputs).
- **Sign-in:** Supabase.
- **Live Waze feed:** Upstash Redis.
- **ML:** Python scripts, refreshed by scheduled jobs.
- **Data window:**
  - Toll data: 2022–2025 is real. January–June 2026 is synthetic, made by the team.
  - The record ends 30 June 2026; uploads dated after that are held back.
  - Live Waze data runs from 4 August 2026.
- **Analytics pages:** Traffic, Incidents and Emissions each have Descriptive, Predictive and Prescriptive tabs.

Role	Pages it can open
data-analyst	Everything
tcc-operator	Everything except Emissions, Data Management and Audit Log
incident-operator	Same as tcc-operator, and also no Scenario Sandbox. This is also the fallback when a role is missing.

There is no "admin" role; "Admin" is only the name of a sidebar group.

2. Dashboard shell

- **Sidebar groups:**
  - Analytics: Overview, Traffic, Incidents, Emissions
  - Operations: Live Map, Maintenance, Mobile App
  - Planning: Scenario Sandbox
  - Admin: Data Management, Audit Log
- **Sidebar behaviour:** collapses to an icon rail, and becomes a drawer below 980 px.

- **Top bar:**
  - Breadcrumb.
  - Effects toggle: moving clouds, sun glow, grid lines.
  - Theme: Light, Dark or System.
  - Live clock.
- **Guards:**
  - A password-only session that hasn't passed the authenticator step is dropped.
  - A page your role can't open shows a “Not available for your role” card, then sends you to Overview after 5 s.
- **Idle sign-out:** after 5–60 minutes (default 15), with a one-minute warning, shared across tabs.
- **Audit:** page views and logouts are logged.
- **Log out:** a confirmation dialog with the mascot.

3. Sign-in and account

- **Sign-in scene:**
  - A 3D toll-plaza intro with Enter SmartFlow, Skip intro and Replay. It shows no data.
  - Plays once per browser session.
  - Falls back to a still image when 3D fails, effects are off, or reduced motion is set.
- **Two-step sign-in:**
  - Password, then a 6-digit authenticator code.
  - Accounts without an authenticator set one up on the spot, with a QR code or a setup key.
  - The backend refuses any token that hasn't passed the code step.
- **Remember this device for 7 days:**
  - Skips only the code; the password is still needed.
  - Revoked by a password change or by “Sign out of all devices”.
  - Shown as unavailable when the server has no device-trust secret.
- **Password reset by email:**
  - A link sent from Navie's mailbox. The reply is the same whether or not the account exists.
  - Rate limited, with a 30-day cool-down between changes.
  - The authenticator code is still asked for after the link.
- **Remember email:** a half-finished sign-in resumes at the code step.

Settings page

- Display name and idle timeout.
- Password change; on a remembered device, the authenticator code is needed first.
- Forget this device, Forget email, Sign out of all devices.
- Theme and effects.

4. Overview

Everything on this page comes from the live Waze feed (`/api/map-comparison/real-time`), refreshed every 60 s. A jam level of 3 or more, or a speed under 10 km/h, counts as congested.

Hero

- A headline worked out from the counts: “N exit-directions congested”, “N slow” or “Corridor flowing”.
- The slowest reading on the corridor.

- How fresh the feed is: Live, Feed stale, or unreachable.
- Buttons: Open Live Map, Traffic forecast.

Navie

- The 3D car greets you with typed text: an intro on first visit, “welcome back” later, plus a live-feed line.
- Clicking it gives jokes, tips and time-of-day lines.
- These are fixed lines, not AI.

Right now

- Congested, slow and clear count cards.
- A “Slow and congested now” list (top 3, with See all) showing queue, delay, speed, km and direction.

Live Corridor Status

- Northbound and southbound strips with jam bands per exit, plus queue and delay facts.
- Clicking an exit opens it on the Live Map.

5. Traffic

Filter bar (Descriptive tab only)

- **Range:** 3 mo, 12 mo, All, or Custom dates. Every preset ends on the last day of data.
- **Weather:** All, Dry or Wet. An hour is wet when average rainfall across stations is above 0.3 mm.
- **Class:** a dropdown for Class 1, 2 or 3. It changes every chart.
- **Filter chips:** Range always shows its exact dates; Weather and Class appear only when changed, with an × to reset.

Descriptive

KPIs

- Total Volume, with change vs the previous equal period.
- Avg Daily Volume.
- Peak Hour (weekdays).
- Busiest Plaza.
- Congestion Index: Waze jam level on a 1–5 scale.

Volume Trend

- Hourly (only for spans of 14 days or less), Daily, Weekly or Monthly.
- A northbound/southbound split.
- Click a point for its detail.
- Insight: the busiest, quietest and median day.

Volume by Plaza

- The top 10 plazas plus an “Others” bar.
- SCTEX entries are counted at Mabiga.
- “View all plazas” opens a sortable table.

Holiday vs Normal Days / Arena Event Impact

- Shows the 8 days furthest from normal, as bars above or below the baseline.
- The baseline is the same weekday within ±45 days, excluding events and holidays, with at least 4 control days.
- Events can be measured at the Venue (CDV/PH Arena), Bocaue Interchange, Tambubong, Marilao, or all arena exits combined.
- Event names are standardised when read.
- View all: a sortable table with a year filter.

Average Volume by Hour and Day

- A heatmap with a colour key you can drag along to highlight cells.

Average Speed in Jams by Hour

- Waze speed and jam level, with the slowest hour.
- Filters by date only.

Predictive

Volume Forecast

- **Models:** Prophet, Holt-Winters, Holt's Linear, SARIMAX and LSTM. The champion is the top accepted model.
- **Bands on the chart:** Past (training), Present (held out), Future.

- **Controls:**
  - Models.
  - Range: trims the history shown.
  - View: Daily, Weekly or Monthly.
  - Horizon: 2 wk to 3 mo.
  - Weather switch: adds rainfall bars and swaps to weather-free versions of the models.
- **Summary column:**
  - Forecast daily average.
  - Typical error for the chosen horizon.
  - The selected model.
  - Validation Evidence: WMAPE, MASE, scored days, period.
  - The separate 48-hour model.
- **Validation details:** the full metrics table, the 48-hour model comparison, and the weather evidence.
- **Hourly drill-down:** click a day to see a Day Summary of peak and quiet hours, predicted vs actual, and the day's weather.
- **48-hour plaza-hour model:** drawn as an orange dashed line.
- **Generate Report:** opens an AI narrative.

Predictive Congestion State Map

- **Header:** the time it was issued, the model badge, and a Refresh-overdue badge.
- **Range:** Next 12 h, 24 h or 7 days.
- **Heatmap:** exits by hour, coloured Moving, Heavy (10–20 km/h) or Severe (under 10 km/h).
- **Exit picker:** a "5 of 20 exits shown" dropdown.
- **Bar row:** how many exits are congested each hour.
- **Priority episode:**
  - The worst upcoming run, with queue, delay, duration, confidence, and traffic compared with usual.
  - "View all N episodes" opens the full list.
- **Validation evidence strip:** opens 8 blocks: how it was tested, replay, calibration, trust per colour, accuracy by hours ahead, queue and delay accuracy, volume alignment, live score.
- **Narrative Explanation strip:** with Generate Report.

Event Surge Impact by Exit

- Past events, or a specific upcoming date.
- Extra vehicles per exit, counting only exits with a clear rise.
- Held-out error: 11.4% per exit, against 13.7% with no event adjustment.

Prescriptive

Staffing Priority

- Each plaza-hour's forecast compared with its usual level (median) and usual peak (90th percentile).
- The chance of passing the peak, from replaying held-out days.
- Views: Week ahead, or Hour by hour.

Congestion Response Advisory

- The chance of severe congestion over the next 12 h, against an "Act" threshold tuned on archived forecasts.
- Precision and recall, and action cards.
- Copy advisory, and a link to the Mobile App page.

Event Intervention Ranking

- Exits ranked by a lower-bound uplift.

- Plan for an upcoming date, or a typical event day.

6. Incidents

Filters

- Range on every tab.
- Weather (All, Dry, Wet) and type (All, Accidents, Breakdowns) on Descriptive only.

Descriptive

KPIs

- Total Incidents.
- Injuries and fatalities.
- Avg Response Time (breakdowns only).
- Avg Time to Clear.
- Top Hotspot.
- Crash Rate in Rain: adjusted for hour of day, with a 95% interval.

Charts

- **Incident Trend:** accidents and breakdowns by day, week or month.
- **Hotspots by Km Segment:** top 10, with a sortable View all.
- **Causes:** a toggle between accident causes, breakdown causes and accident types.
- **Dry vs Wet: Incidents per Day:** adjusted for how many dry and wet hours there were, with its own Day and Hour filters.
- **Incidents by Hour and Day:** heatmap.
- **Response Time Breakdown:** by cause or by service. Only about 31% of breakdowns have deployment records.

Predictive

Incident Walk-Forward Forecast

- **Models:** XGBoost, Random Forest, Poisson GLM, Negative Binomial GLM, SARIMAX, LSTM and GRU.
- **Champion:** Negative Binomial GLM. Only models with MASE ≤ 1 qualify, then the highest R^2 wins.
- **Controls:** View, Ahead (2 wk to 3 mo), and Volume and Weather switches that swap to the twin model fits.
- **Accident / Breakdown split:** accidents have their own model (Poisson GLM); breakdowns are the total minus accidents.
- **Summary column:**
  - Forecast incidents per day (114/day now) and the busiest day.
  - Typical error at the horizon, with a warning when it's beyond what the champion clears.
  - Validation Evidence.
- **Drill-down:** clicking a day opens the hourly page.

Predicted Incident Hotspots

- Spatial LSTM per exit plus GWR.
- Horizons of 1, 7, 14 or 28 days.
- A risk strip with four bands.

Expected Breakdown Response Time

- Cox model vs XGBoost.
- Filtered by day and 3-hour block, grouped by cause or by service.

Hourly drill-down page

- Hourly bars with each model's curve; the curve is the daily total spread over a typical day's shape.
- Rain and temperature, Weather chips.
- An hour-by-source heatmap.

Prescriptive

- **Resource Staging & Patrol Repositioning:** places units busiest-first, showing the cumulative share of incidents covered.
- **Patrol Alert Schedule:** hours whose Poisson lower bound is above the weekday average.
- **Clearance Response Protocol:**
  - A survival curve (Kaplan–Meier) of clearance times, with checkpoints at P25, P50 and P80.
  - An escalation rule: if still open at P50, the chance it runs past P80.
  - Red/amber/green against targets: 30/45 min for accidents, 10/18 min for breakdowns.

7. Emissions

How CO_2 is worked out

- CO_2 = vehicle-km × EMEP/EEA 2023 factors: 196, 689 and 798 g/km for Class 1, 2 and 3.
- Vehicle-km come from real trip distances.
- These are modelled figures, not measurements.
- The factors describe an EU fleet, not the Philippine fleet.

Descriptive

KPIs

- Total CO_2, with a low–high range.
- Avg Daily CO_2.
- Heavy-Vehicle Impact.
- Peak Emission Hour.
- Delay-Induced CO_2: idling in lane-blocking queues.
- Estimated Air Quality: modelled AQI out of 5.

Cards

- CO_2 Trend by Class.
- When Emissions Happen: by hour or by day.
- Fleet Mix vs Pollution Load: shares of CO_2, NOx and PM2.5.
- Heavy-Vehicle Share of CO_2.
- Air Quality by Month.
- CO_2 by Hour and Weekday.

Predictive

Corridor CO_2 Walk-Forward Forecast

- **Models:** Gradient Boosting (champion: WMAPE 5.6%, MASE 0.98), Polynomial, LSTM, and Volume-derived.
- **Validation:** 42 rolling origins, 7 days ahead.
- **Ahead:** only 7 days and 2 weeks are offered, because longer horizons lose to repeating last week.
- **Summary column:** 2,301 t CO_2/day.

Forecasted Fleet Composition

- Models: Dirichlet (champion), XGBoost, LSTM and Prophet.
- The heavy-vehicle share, observed against projected.

Prescriptive

- **Clear blockages as fast as the fastest 10%:**
  - Tonnes saved looking back, with a bootstrap range.
  - Tonnes saved ahead, using the incident forecast.
  - Where and when the savings fall.
- **Class 3 window advisory:** the forecast's peak-congestion hours. It claims no CO_2 saving.
- **Tolling recommendation:** the fewest exits that together carry at least 80% of Class 3 CO_2.

Three recommendations were removed on 2 Nov 2026 because they weren't backed by data: the share shift, the responder deployment, and the TOPSIS ranking.

8. Live Map

- **Left map, Waze Real-Time:** live jams and alerts, with Active Reports and Avg Speed.
- **Right map, Forecasted Traffic:** the congestion forecast up to 336 h ahead, with a model card.
- **Shared controls:**
  - Jump to exit, 3D tilt, Expand.
  - A horizon timeline with daily peaks.
  - A corridor strip that switches between Live and Forecast.
  - A closures overlay from Maintenance.
- **When the database is down:** sample data is shown.

9. Scenario Sandbox

An in-browser simulation of individual vehicles.

- **Setup:**
  - Carriageway: Both, northbound or southbound.
  - Forecast day: seeds the inflow and fleet mix from the forecasts.
- **Tiles:** agents, speed, throughput, longest queue, CO_2 rate, density.
- **Simulation tools:**
  - Corridor and lane setup.
  - Interventions: lane closures, speed limit, lane reallocation, booths.
  - Scenario events: breakdowns, collisions, flood, roadworks, rain. Their durations come from calibration data.
  - Baseline comparison.
  - A confidence run: 10 runs, shown with a 95% interval.
  - Replay.
- **Recommendation panel:** rule-based, not AI.
- **SmartFlow Copilot:**
  - Typed commands in English, Tagalog or Taglish, turned into a plan of actions by an AI model (GLM).
  - The backend checks the plan; the operator reviews it and applies it.
  - Only data-analysts and tcc-operators can use it.

10. Mobile App (Mobile Control Centre)

- **Tabs and sections:** each of the five app tabs has sections you can switch on or off.
- **Phone preview** of the result.
- **Advisories:** up to 6, each Info, Warning or Critical.
- **Saving:** a save bar with validation.
- **Storage:** saved to `nlex_mobile_config`. The app reads it from the public `/api/mobile-config`.

11. Maintenance

- **Scheduling:** km 12–88.25, lanes, and times.
- **Status steps:** scheduled → in progress → completed or cancelled.
- **Checks:**
  - Overlapping lanes are blocked unless you choose “Schedule anyway”.

    - A start time in the past is refused.
- **Views:** Table, Timeline and Corridor.
- **Public list:** the list is public for the mobile app.

12. Data Management

- **Upload:**
    - CSV, TSV, JSON or Excel files.
    - Options: Check only, or Upload & load.
    - Five checks: parse, classify, clean, transform, load.
- **Where it lands:**
    - Toll files replace plaza-hours across the traffic and emissions tables.
    - Incident and emissions files go to bronze, then silver.
- **History:** recent uploads with Undo.
    - Undo works for the last 10 loads.
    - It is blocked if a later load touched the same tables.
- **Also shown:** the accepted formats table and the weekly retrain status.

13. Audit Log

- **What's logged:** sign-in and settings events, page views, uploads, maintenance changes, mobile-config saves, and model refreshes.
- **Insights:** over 7, 30 or 90 days: usage, maintenance turnaround, uploads, retrains.
- **Filters:** search, Category, Severity, Date.
- **Table and export:** a sortable table and Export JSON.

14. Backend platform

- **Rate limit:** 200 requests per 15 minutes per IP.
- **Caching:**
    - Live data 30 s, forecasts 60 s, maintenance and audit 15 s, everything else 3 h.
    - Serves the last copy while it refreshes.
    - Signed-in responses are never stored.
- **Database connections:** a pool of 2–10 that replaces itself after repeated timeouts. TLS is on by default but does not verify the server certificate (`relaxed`).
- **Health check:** `/api/health` does not check the database.

15. Data and ML pipeline

Warehouse

- **bronze:** raw copies.
- **silver:** cleaned data: Waze jams, weather, incidents, events.
- **gold:** facts and model outputs (`ml_*`).
- **public:** mostly views and legacy tables.

Models

Model	Champion now	How it's chosen	Validation
Daily volume	Prophet (not confirmed in the stored table)	Must beat the baselines on WMAPE and MASE; then lowest WMAPE	Rolling origins, 14 days ahead
48-hour plaza-hour	A learned model only if it beats both baselines	Lowest WAPE	Tested on all of 2025
Congestion state	XGBoost, 63.9% in the latest log	Beats the baselines; then best at catching severe hours within 0.5 pt of the top accuracy	Last 12 days held out; horizons 1–168 h
Incident counts	Negative Binomial GLM (accidents: Poisson GLM)	MASE ≤ 1, then highest R^2	294-day holdout
CO_2	Gradient Boosting	WMAPE beats the baseline and MASE < 1	42 origins, 7 days ahead
Fleet mix	Dirichlet	Aitchison distance and skill	42 origins
Event surge	Per-exit uplift	—	33 held-out events

Other incident models: severity (XGBoost), spatial (GWR and Spatial LSTM), breakdown response (Cox), and weather–speed (XGBoost).

Schedules

- **Congestion refresh:** hourly, via a Windows task and a GitHub workflow.
- **Weather:** hourly.
- **Weekly retrain:** every model group, Sunday 22:00, with backup and rollback.

16. Known gaps and stale bits

1. > **Roles can be edited by the user.** The role is read from Supabase `user_metadata`, which users can change themselves. Sign-up is also open. This is a real security gap.
2. **The congestion badge doesn't match the latest run.** The card shows "Stacked LSTM · 64.9%", but the latest hourly log names XGBoost (63.9%). This PC's hourly run also skips the LSTM, GRU and dual-path models because TensorFlow isn't installed, which fits a second machine writing to the same table.
3. **January–June 2026 traffic is synthetic.** The manuscript needs to say so.
4. **Code that no page uses:**
   - `/api/dashboard`, the corridor-status service, the AI Explain service, and the old AI-sandbox simulate routes.

- The `data-management` route isn't mounted.
- Several incident model panels aren't on any page.

5. **Small bugs:**
   - The Volume Forecast's hourly fetch is hard-coded to `localhost:4000`.
   - Its fetch errors leave the card stuck on loading.
   - The 48-hour model's stale flag is never shown.
   - Live Map shows “Toll Plazas 20” and “NLEX Exits 20” as hard-coded numbers.
6. **Stale text:**
   - The upload dialog says it “cannot undo a load”, but Undo exists.
   - Several old comments: the incident apportionment source, the emissions script header, and the congestion README's figures.
7. **Missing UI hints:** incident-operators can open Maintenance and Mobile App, but their saves get a 403 with no hint in the UI.

SmartFlow NLEX · System inventory generated 10 November 2026 from the working tree on branch Main-Dashboard.